

Description of ICs

CS 224

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7404 : Hex 1-Input NOT (Inverter)

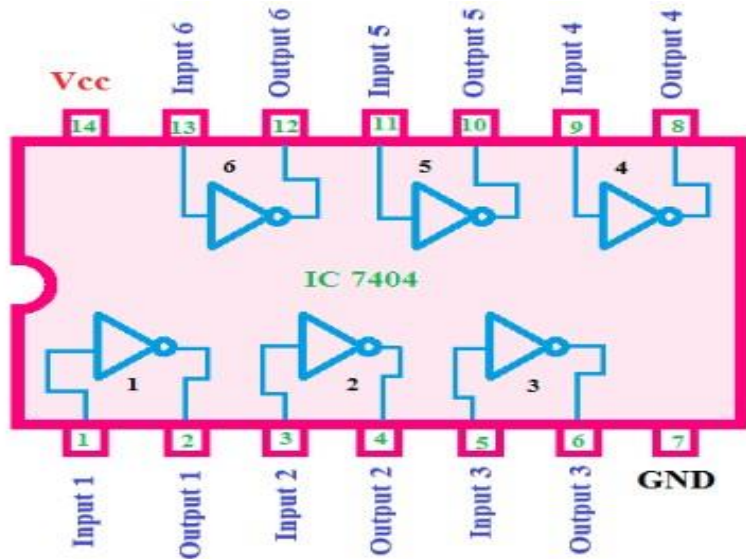


Fig 1 :Pin Diagram of IC 7404

Input (X)	Output (X')
0	1
1	0

Fig 2 :Truth Table of NOT gate

- The IC 7404 contains six NOT gates (Pin diagram shown in Figure 1)
- The six gates are numbered like 1,2,3,4,5 and 6 in the Pin diagram.
- Pin X is the input to Xth NOT gate, where its output can be obtained from Pin X+1, where $X=\{1,3,5,9,11,13\}$
- Pin 14 is the positive voltage supply (Vcc) and Pin 7 is Ground (Gnd).

7408 : Quad 2-Input AND gate

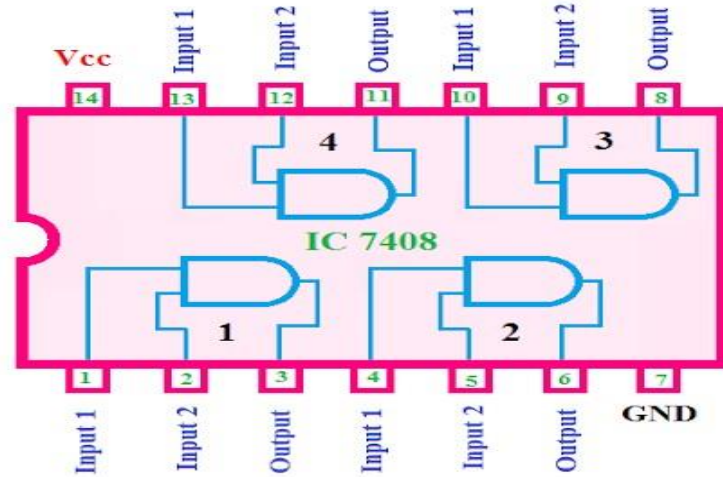


Fig 1 :Pin Diagram of IC 7408

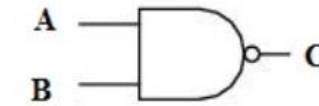
Input 1 (A)	Input 2 (B)	Output
0	0	0
0	1	0
1	0	0
1	1	1

Fig 2 :Truth Table of AND gate

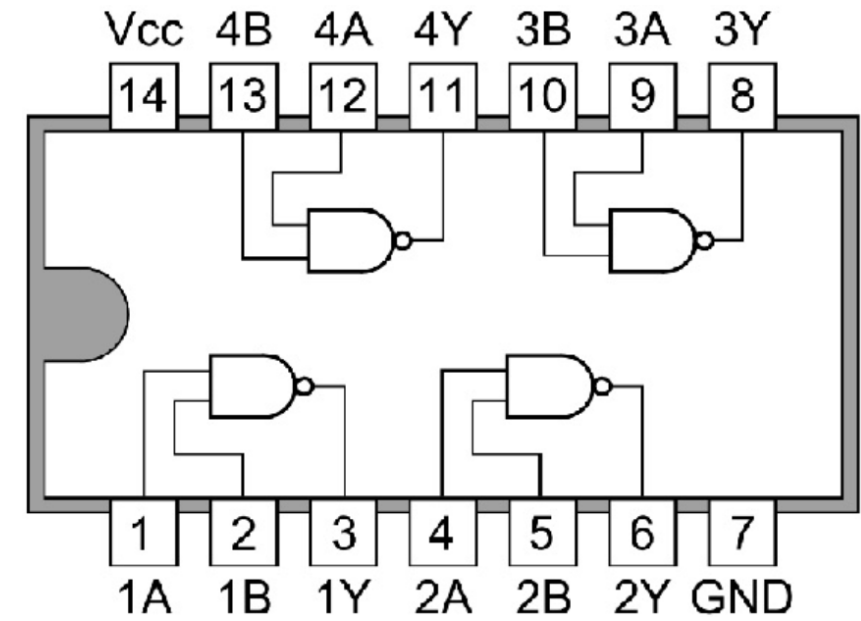
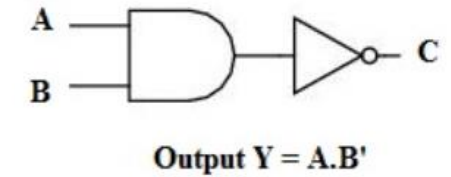
- The IC 7408 contains 4 AND gates (Pin diagram shown in Figure 1)
- The four AND gates are numbered like 1,2,3 and 4 in the Pin diagram.
- Pin 1 and Pin 2 are inputs to the 1st AND gate, where its output can be obtained at Pin 3. Similarly, Pin 4,5 are inputs and Pin 6 is the output of gate 2 and so on.
- Pin 14 is Vcc and Pin 7 is ground (Gnd).

7400 : Quad 2-Input NAND

- Pin1: It is an A-input Gate-1
- Pin2: It is B-input Gate-1
- Pin3: It is Y-output Gate-1
- Pin4: It is an A-input Gate-2
- Pin5: It is a B-input Gate-2
- Pin6: It is a Y-output Gate-2
- Pin7: It is a GND terminal
- Pin8: It is a Y-output Gate-3
- Pin9: It is a B-input Gate-3
- Pin10: It is an A-input Gate-3
- Pin11: It is a Y-output Gate-4
- Pin12: It is a B-input Gate-4
- Pin13: It is an A-input Gate-4
- Pin14: It is a Vcc pin (Positive Supply)

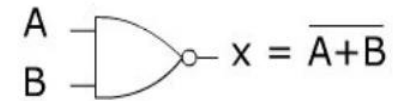


A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

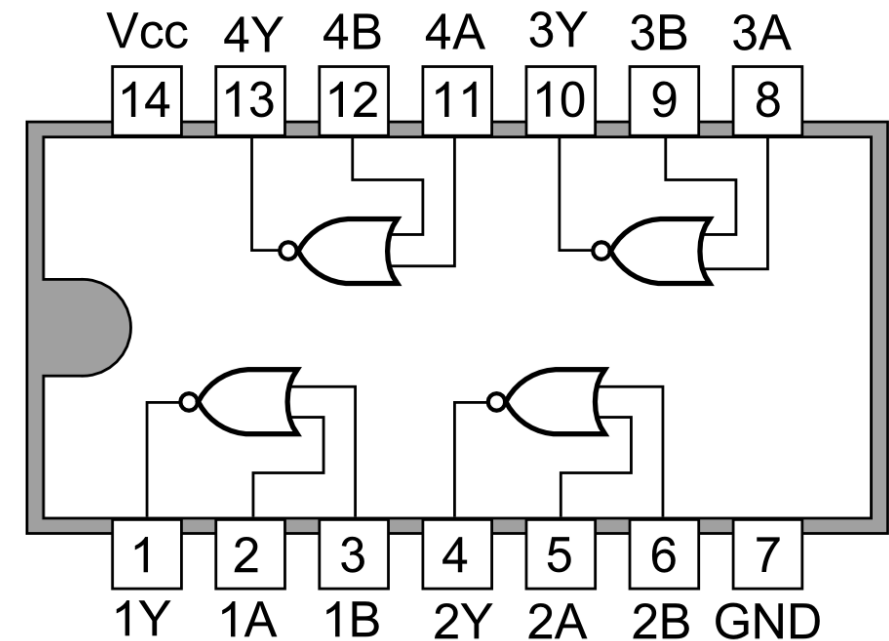


7402 : Quad 2-Input NOR

Pin Number	Description
1	Y Output Gate 1
2	A Input Gate 1
3	B Input Gate 1
4	Y Output Gate 2
5	A Input Gate 2
6	B Input Gate 2
7	Ground
8	A Input Gate 3
9	B Input Gate 3
10	Y Output Gate 3
11	A Input Gate 4
12	B Input Gate 4
13	Y Output Gate 4
14	Vcc - Positive Supply



A	B	$x = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



7432 : Quad 2-Input OR Gates

- **74LS32N** has 12 input output pins for a total of four OR gates.

- This IC power on using the Vcc (14) and ground(7) pins.

- The operating voltage of the IC is +5V, but it can also be operated in +7V.

- The output voltage of the IC on the pin Y will be equal to the operating voltage of the IC.

- As per the **OR gate truth table**, when either one or both the input of the Gate is high the output will be high.

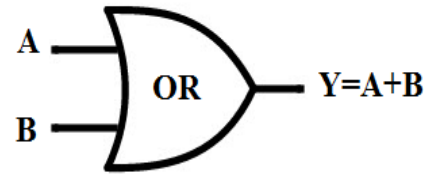


Fig 1 : OR Gate & Truth Table

Inputs		Output
A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

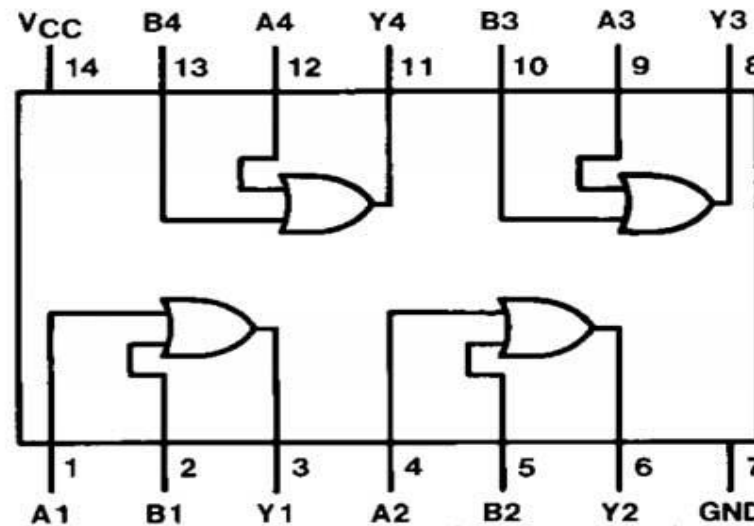


Fig 2 : Pin Layout



Fig 3 : Pin Description



7486 : Quad 2-Input XOR Gates

- The **7486** is a high-speed CMOS Logic Quad **2-input XOR Gate**.
- 7486 has 14 pins in total four groups of exclusive OR gates.
- The output of the XOR gate is 1 when both inputs (A, B) are different, and is 0 when both the inputs are the same.
- Operating Voltage of IC is +5V.



Fig 1 : IC 7486 XOR/EX-OR



Fig 2 : EX-OR Gate & Truth Table

Inputs		Output
A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

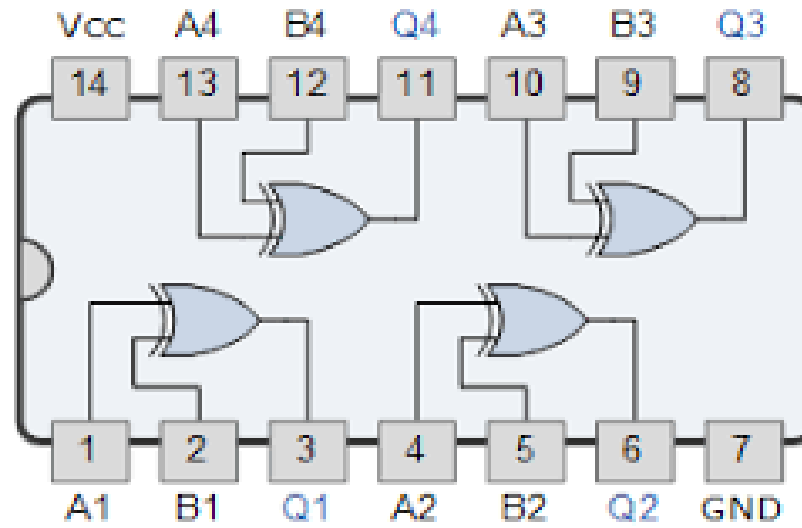


Fig 3 : Pin Layout

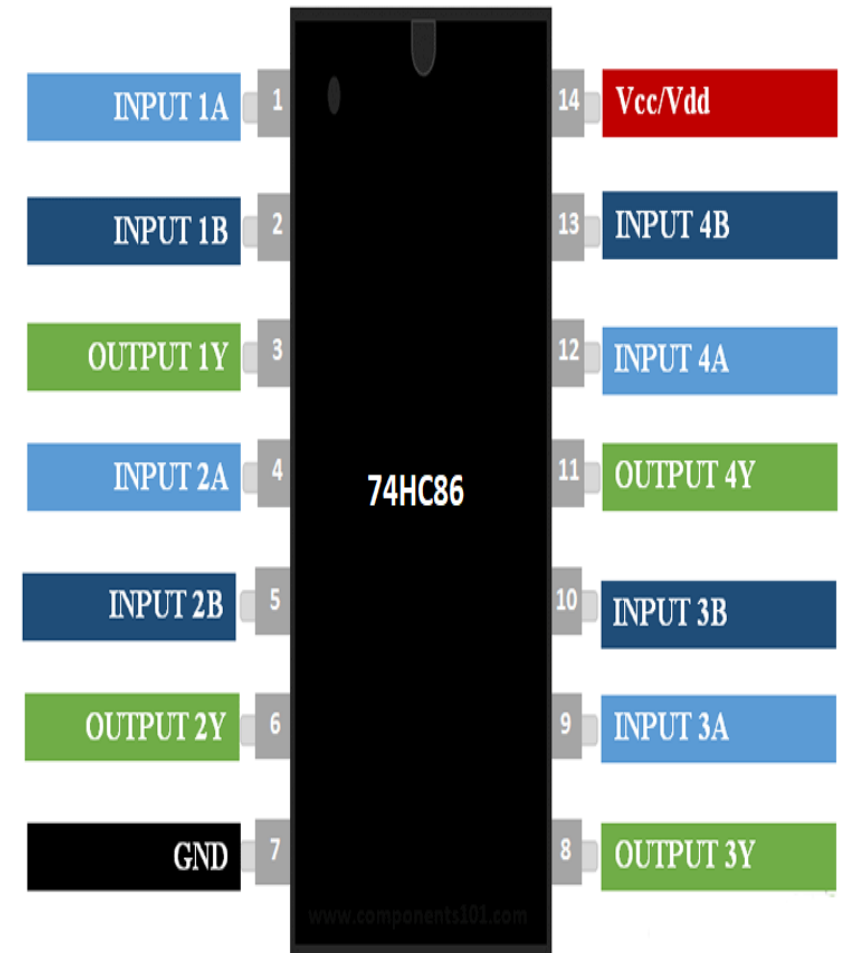


Fig 4 : Pin Description

7410 : Triple 3-Input NAND

- IC 7410 contains three 3-Input NAND Gates
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Fig 1 : Truth Table

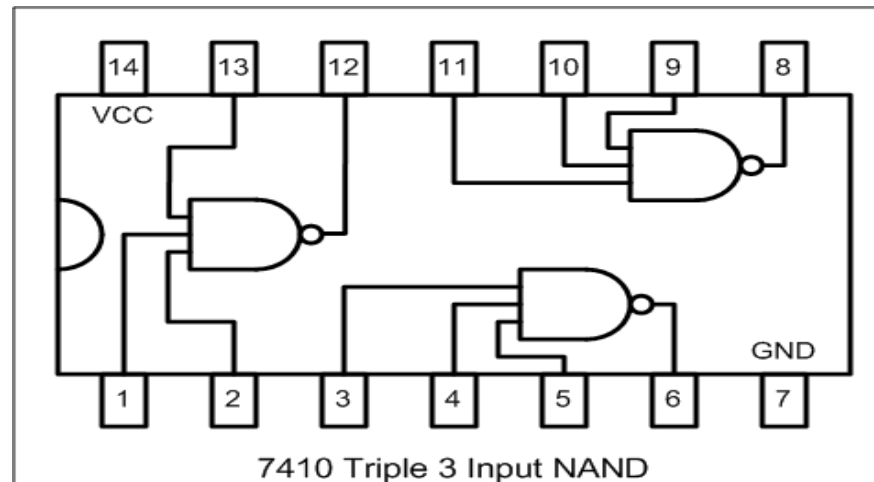


Fig 2 : Pin Layout

Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	A Input Gate 2
4	B Input Gate 2
5	C Input Gate 2
6	Y Output Gate 2
7	Ground
8	Y Output Gate 3
9	A Input Gate 3
10	B Input Gate 3
11	C Input Gate 3
12	Y Output Gate 1
13	C Input Gate 1
14	Vcc - Positive Supply

Fig 3 : Pin Description

7411 : Triple 3-input AND

- IC 7411 contains three 3-Input AND Gates
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Fig 1 : Truth Table

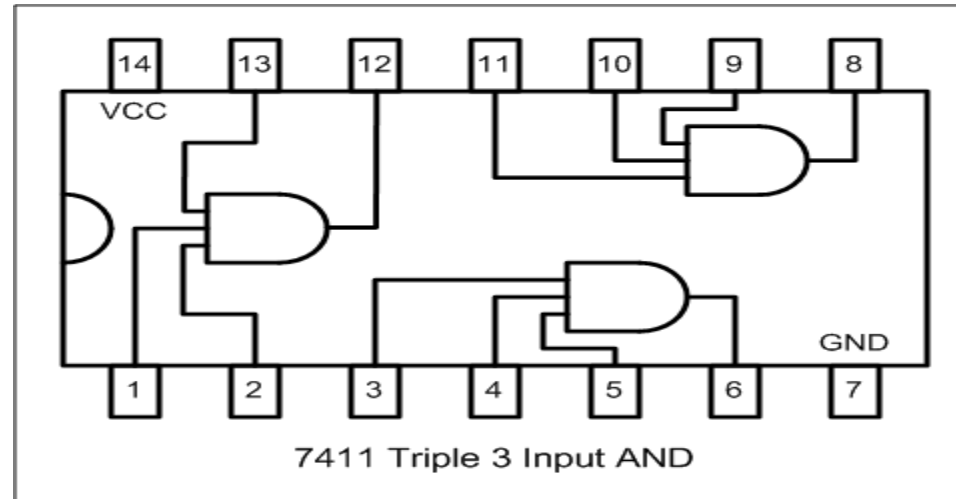


Fig 2 : Pin Layout

Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	A Input Gate 2
4	B Input Gate 2
5	C Input Gate 2
6	Y Output Gate 2
7	Ground
8	Y Output Gate 3
9	A Input Gate 3
10	B Input Gate 3
11	C Input Gate 3
12	Y Output Gate 1
13	C Input Gate 1
14	Vcc - Positive Supply

Fig 3 : Pin Description

7427 : Triple 3-Input NOR

- The function of a 3-input NOR is given by

$$Y = \overline{A + B + C}$$

- Three independent 3-Input NOR Gates
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range
- Wide Operating Conditions

Input	Input	Input	Output
A	B	C	Y
0	0	0	1
X	X	1	0
X	1	X	0
1	X	X	0

Where X = Either 1 or 0

Fig 1 : Truth Table

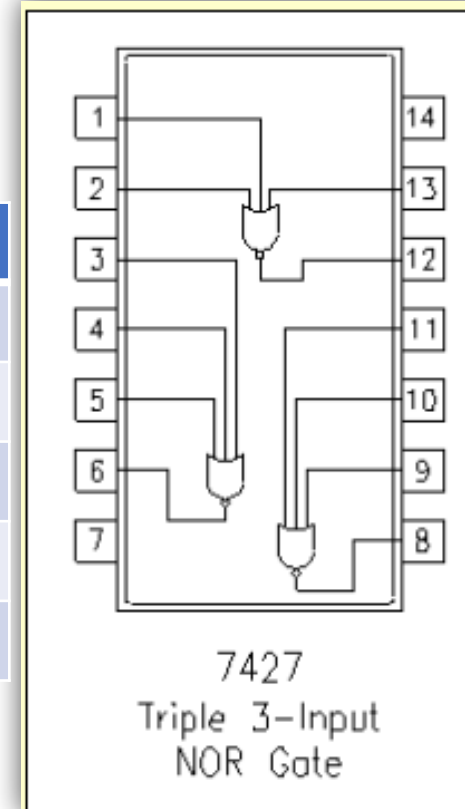


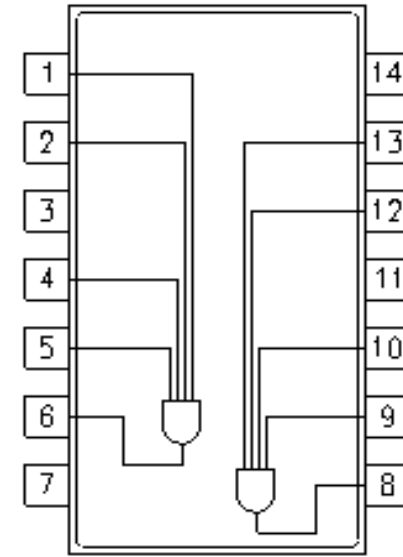
Fig 2 : Pin Layout

Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	A Input Gate 2
4	B Input Gate 2
5	C Input Gate 2
6	Y Output Gate 2
7	Ground
8	Y Output Gate 3
9	A Input Gate 3
10	B Input Gate 3
11	C Input Gate 3
12	Y Output Gate 1
13	C Input Gate 1
14	Vcc - Positive Supply

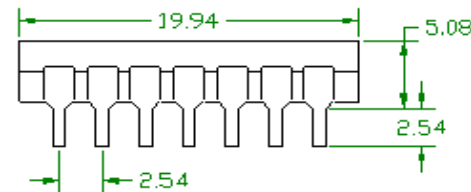
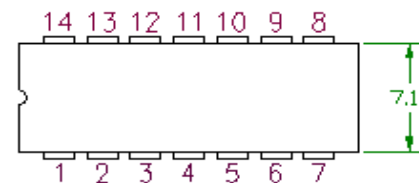
Fig 3 : Pin Description

7421 : 4-Input AND

- Two 4-Input AND Gates
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range
- Wide Operating Conditions



7421
Dual 4-Input
AND Gate

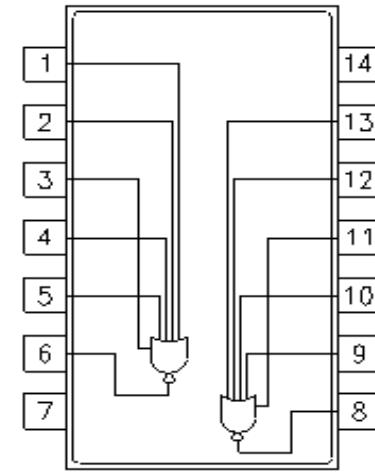


14-Pin DIP

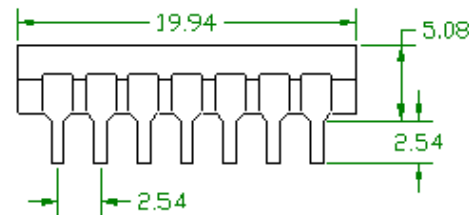
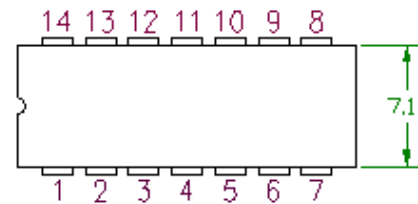
Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	Not Connected
4	C Input Gate 1
5	D Input Gate 1
6	Y Output Gate 1
7	Ground
8	Y Output Gate 2
9	A Input Gate 2
10	B Input Gate 2
11	Not Connected
12	C Input Gate 2
13	D Input Gate 2
14	Vcc - Positive Supply

7425 : Dual 4-Input NOR

- Two 4-Input NOR Gates
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range
- Wide Operating Conditions



7425
Dual 4-Input
NOR Gate with Strobe



14-Pin DIP

Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	Strobe Gate 1
4	C Input Gate 1
5	D Input Gate 1
6	Y Output Gate 1
7	Ground
8	Y Output Gate 2
9	A Input Gate 2
10	B Input Gate 2
11	Strobe Gate 2
12	C Input Gate 2
13	D Input Gate 2
14	Vcc - Positive Supply

7420 : Dual 4-Input NAND

- The function of a 4-input NAND is given by $Y = \overline{ABCD}$
- This device contains two independent gates each of which performs the logic NAND function
- Outputs Directly Interface to CMOS, NMOS and TTL
- Large Operating Voltage Range
- Wide Operating Conditions

Input	Input	Input	Input	Output
A	B	C	D	Y
X	X	X	0	1
X	X	0	X	1
X	0	X	X	1
0	X	X	X	1
1	1	1	1	0

Where X = Either 1 or 0

Fig 1 : Truth Table

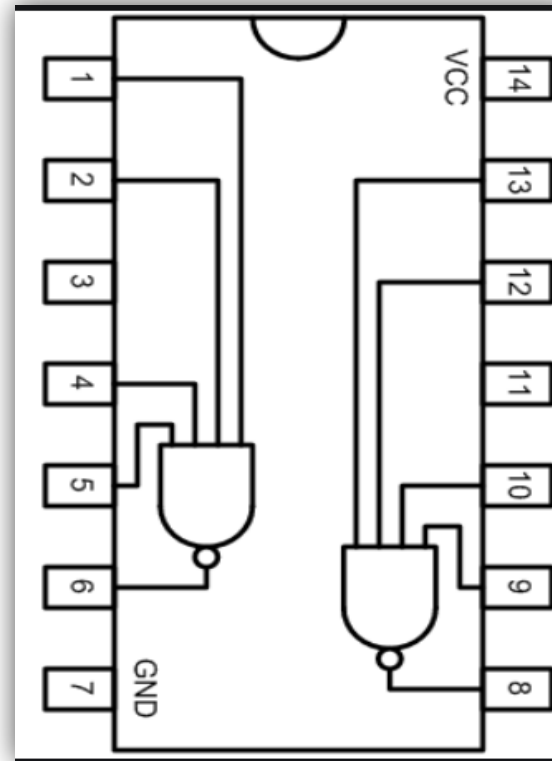


Fig 2 : Pin Layout

Pin Number	Description
1	A Input Gate 1
2	B Input Gate 1
3	Not Connected
4	C Input Gate 1
5	D Input Gate 1
6	Y Output Gate 1
7	Ground
8	Y Output Gate 2
9	A Input Gate 2
10	B Input Gate 2
11	Not Connected
12	C Input Gate 2
13	D Input Gate 2
14	Vcc - Positive Supply

Fig 3 : Pin Description

74151 - 8:1 MUX

If MUX has number 151 or 151P then 5=Y, 6= $\sim$ Y
If 151N then 5= $\sim$ Y and 6=Y

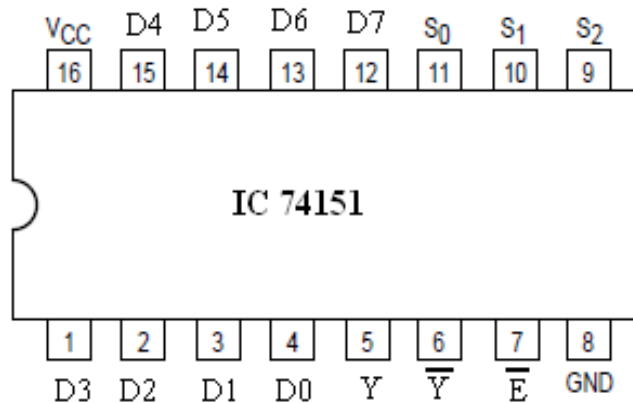


Fig A: Pin Diagram 74151 (8:1 MUX)

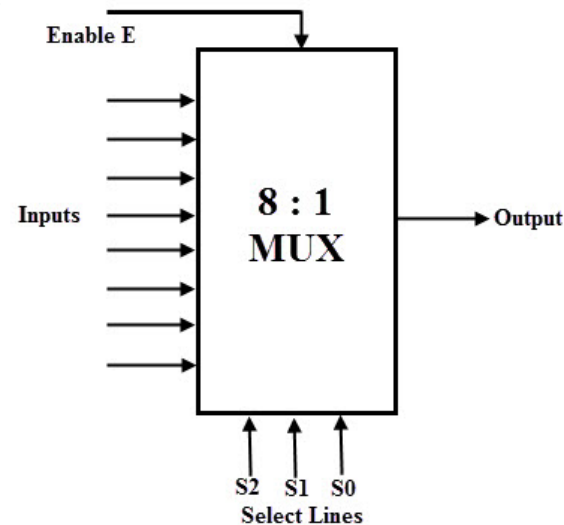


Fig B: Block Diagram of 8:1 MUX

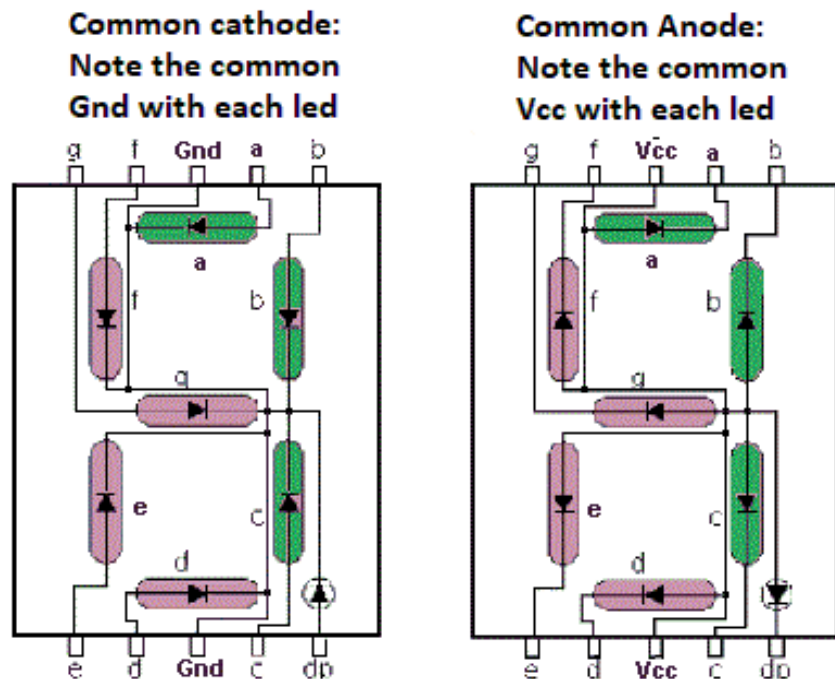
Select Data Inputs			Output
S ₂	S ₁	S ₀	Y
0	0	0	D ₀
0	0	1	D ₁
0	1	0	D ₂
0	1	1	D ₃
1	0	0	D ₄
1	0	1	D ₅
1	1	0	D ₆
1	1	1	D ₇

Fig C: Truth Table of 8:1 MUX

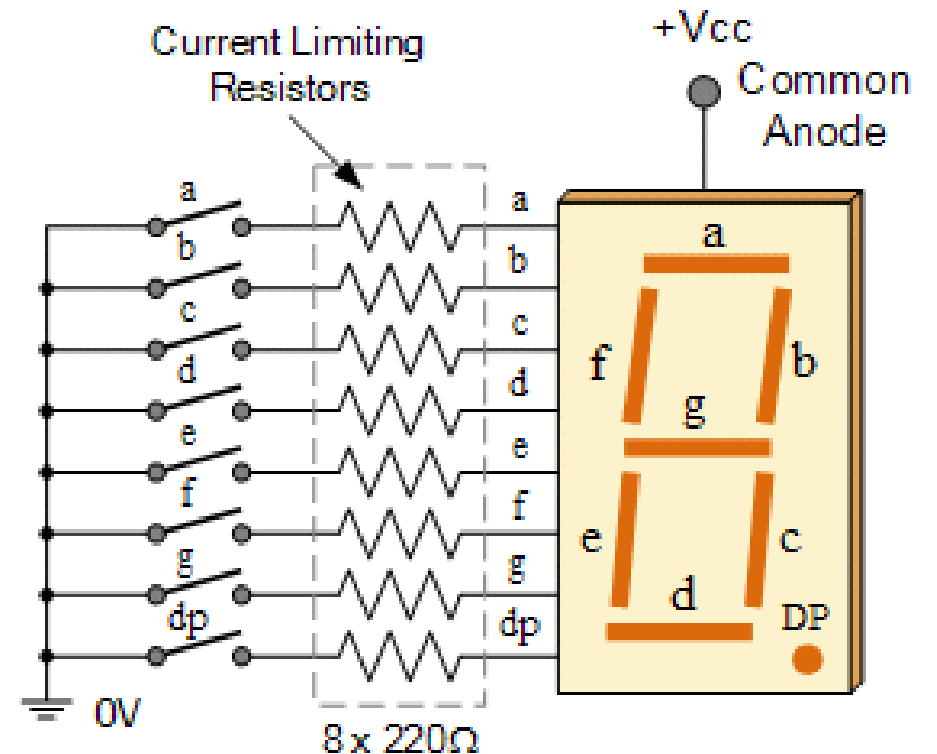
- IC 74151 represents 8:1 Multiplexer
- Pins D0 (Pin 4), D1 (Pin 3), ..., D7 (Pin 12) in Fig A denote the 8 inputs to the MUX.
- Pin 7 is the active low enable to the IC.
- S0 (Pin 11), S1 (Pin 10) and S2 (Pin 9) are three select lines.
- Outputs can be derived from Pin 5 (Uncomplemented, Y) and Pin 6 (Complemented, $\bar{Y}$)
- Pin 16 is Vcc and Pin 8 is Ground (GND).

7-segment LED display

- There are two types:
 - Common Anode – middle pins are connected to Vcc - number 5501
 - Common Cathode – middle pins are connected to Gnd – number 5503
- Each LED has to be connected to power supply via a current limiting resistor – use 3.5 K-Ohm for common Cathode
- Use 1.5 K-Ohm and 680 Ohm for common Anode



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BCD to 7-segment decoder – IC 7447

- The BCD inputs is DCBA with "D" as the MSB
 - Outputs are a-f going to respective pins of the 7-segment display
 - The other pins are special pins. Connect the pins 3,4,5 to VCC
 - This gives common ANODE output. So use appropriate LED display. Use less resistance
-
- Pin-3 is Lamp-Test-bar. If you connect this to "0" then all LEDs glow irrespective of DCBA
 - Pin-4 is BI-bar = Blanking Input. If this is "0" then all LEDs are off irrespective of DCBA
 - RBI-bar = Ripple Blanking Input. When this is "0" and DCBA=0000 the display will blank otherwise the number is displayed. This is used if you want to connect more than one digit of BCD. The MSB of DCBA from one display is connected to the RBI-bar of next display. This is used to blank the display of higher digit if we don't want to display zero. For example "09" is seen as "_9"

